

Deep Sanbui

B.Tech Undergraduate, Computer Science & Engineering
Heritage Institute of Technology, Kolkata

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Academic Qualifications

Degree/Certificate	Institute	CGPA/%
B.Tech	Heritage Institute of Technology, Kolkata	9.476/10
WBCHSE (XII)	Bargachia Union Priyanath Pathsala, Howrah	90%
WBBSE (X)	Pantihal Damodar Institution, Howrah	91.28%

Technical Skills

- **Languages:** Python, C, SQL, Java, JavaScript
- **Web Development:** MongoDB, Express.js, React.js, Node.js, Tailwind CSS, REST APIs
- **AI & ML:** Scikit-learn, Pandas, NumPy, NLP, RAG, LangChain, LangGraph
- **Concepts:** LLM Integration, Informational Retrieval, Vector Databases, Similarity Search
- **Tools:** VS Code, Git, GitHub, Streamlit, Tavily, Vercel, Render, Cloudinary

Key Projects

- **Multi-Agent Research System** (2026)
Tech Stack: Python, LangChain, Mistral AI, Tavily, Streamlit
 - Designed a multi-agent pipeline with dedicated Search, Reader, Writer, and Critic agents for automated research report generation.
 - Integrated Tavily API for real-time web search and implemented a web scraping agent to extract deeper content from relevant URLs.
 - Coordinated agent workflows using LangChain with Mistral AI as the underlying LLM for reasoning, writing, and critique.
 - Built a Streamlit UI with live step status indicators, tabbed results view, and markdown report download.
- **Smart College Portal** (Ongoing)
Tech Stack: React.js, Node.js, Express.js, MongoDB, Socket.IO, Gemini AI, JWT, Tailwind CSS
 - Built a full-stack role-based academic management system for Admin, HOD, Faculty, and Student users with secure JWT authentication.
 - Developed department-wise student, faculty, and course management modules with interactive dashboards.
 - Integrated Gemini AI to generate automated class routines with faculty-course mapping, preview, and PDF download features.
 - Implemented real-time notifications and announcements using Socket.IO with unread notification tracking and toast alerts.
- **Healthcare Symptom Checker Chatbot** (2026)
Tech Stack: Python, Scikit-learn, Pandas, NumPy
 - Predicted diseases from user-described symptoms using a Random Forest Classifier trained on a structured symptom dataset.
 - Implemented symptom extraction using synonym mapping, exact matching, and fuzzy matching for natural language handling.
 - Designed a guided questioning system that dynamically asks disease-specific follow-up questions to improve prediction accuracy.
- **Library Management System** (2025)
Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, Cloudinary
 - Built a full-stack web application with role-based access control for Admin, Librarian, and Student users.
 - Implemented book management with image uploads, issue/return tracking, and automated fine calculations.
 - Integrated JWT authentication and OTP-based password reset using Nodemailer.
 - Deployed the application using Vercel (frontend) and Render (backend).

Achievements

- Selected for the prestigious **Visteon Scholar Programme** as one of only **30 students** chosen out of approximately 800 students from Heritage Institute of Technology.
- **ChronoSync** was shortlisted in the **Top 50 teams** during college-level screening for **SIH**.